

CSES — Range Queries Problem Reference

All tasks: 1 s, 512 MB. Ranges and array indices are 1-based.

1143 — Hotel Queries

n hotels have free-room counts h_i . For each arriving group requiring r , assign the *first* hotel with at least r rooms, then decrease that hotel's free rooms by r ; output 0 if none exists. **In:** n m ; $h_1 \dots h_n$; $r_1 \dots r_m$
Out: assigned hotel index for every group, or 0
Constraints: $n, m \leq 2 \cdot 10^5$; $h_i, r_i \leq 10^9$.

1144 — Salary Queries

Maintain salaries p_i under point assignments. Query how many employees have salary in the inclusive value interval $[a, b]$. **In:** n q ; salaries; then $!$ k x or $?$ a b
Out: count for each $?$ query
Constraints: $n, q \leq 2 \cdot 10^5$; salaries/query values $\leq 10^9$.

1190 — Subarray Sum Queries

Maintain an integer array under point assignments. After *every* update, report the maximum subarray sum of the entire array; the empty subarray with sum 0 is allowed. **In:** n m ; array; then m updates k x
Out: maximum subarray sum after each update
Constraints: $n, m \leq 2 \cdot 10^5$; $|x_i|, |x| \leq 10^9$. Use 64-bit.

1646 — Static Range Sum Queries

Given a fixed array, answer the sum of each inclusive range $[a, b]$. **In:** n q ; array; then q lines a b
Out: $\sum_{i=a}^b x_i$ for each query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^9$. Use 64-bit.

1647 — Static Range Minimum Queries

Given a fixed array, answer the minimum value in each inclusive range $[a, b]$. **In:** n q ; array; then q lines a b
Out: $\min(x_a, \dots, x_b)$ for each query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^9$.

1648 — Dynamic Range Sum Queries

Maintain an array with point assignment and range-sum queries. **In:** n q ; array; queries 1 k u (set $x_k = u$) or 2 a b (sum)
Out: range sum for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i, u \leq 10^9$. Use 64-bit.

1649 — Dynamic Range Minimum Queries

Maintain an array with point assignment and range-minimum queries. **In:** n q ; array; queries 1 k u (set) or 2 a b (minimum)
Out: range minimum for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i, u \leq 10^9$.

1650 — Range Xor Queries

Given a fixed array, answer the bitwise xor of values in each inclusive range $[a, b]$. **In:** n q ; array; then q lines a b
Out: $x_a \oplus \dots \oplus x_b$ for each query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^9$.

1651 — Range Update Queries

Maintain an array under range additions. Type 1 adds u to every x_i for $i \in [a, b]$; type 2 asks the current value at one position k . **In:** n q ; array; 1 a b u or 2 k
Out: x_k for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i, u \leq 10^9$. Use 64-bit.

1652 — Forest Queries

An $n \times n$ fixed grid contains trees $*$ and empty cells $.$. For each rectangle with corners (y_1, x_1) and (y_2, x_2) , count trees inside it, boundaries included. **In:** n q ; n grid rows; then y_1 x_1 y_2 x_2
Out: tree count for each rectangle
Constraints: $n \leq 1000$, $q \leq 2 \cdot 10^5$.

1664 — Movie Festival Queries

n movies have intervals $[s, e]$. For each arrival/leaving interval $[a, b]$, find the maximum number of whole movies that can be watched: first movie may start at a , consecutive movies may touch ($e_1 \leq s_2$), and last may end at b . **In:** n q ; n movie lines a b ; then q availability lines a b
Out: maximum watchable movies for each query
Constraints: $n, q \leq 2 \cdot 10^5$; all times in $[1, 10^6]$, start $<$ end.

1734 — Distinct Values Queries

Given a fixed array, count the number of distinct values in each range $[a, b]$. **In:** n q ; array; then q lines a b
Out: number of distinct values in each range
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^9$.

1735 — Range Updates and Sums

Maintain an array with three operations: add x to all values in $[a, b]$; set all values in $[a, b]$ to x ; or query the sum of $[a, b]$. **In:** n q ; array; then 1 a b x , 2 a b x , or 3 a b
Out: range sum for each type-3 query
Constraints: $n, q \leq 2 \cdot 10^5$; initial/update values $\leq 10^6$. Use 64-bit.

1736 — Polynomial Queries

Type 1 on $[a, b]$ adds $1, 2, \dots, b - a + 1$ respectively to $x_a, x_{a+1}, \dots, x_b$. Type 2 asks the sum of $[a, b]$. **In:** n q ; array; then 1 a b or 2 a b
Out: range sum for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^6$. Use 64-bit.

1737 — Range Queries and Copies

Maintain a list of versioned arrays, initially containing one array. Operations: set position a in version k to x ; query sum $[a, b]$ in version k ; or append an independent copy of version k as a new version. **In:** n q ; array; then 1 k a x , 2 k a b , or 3 k
Out: range sum for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^3$; $t_i, x \leq 10^9$. Use 64-bit.

1739 — Forest Queries II

Maintain an $n \times n$ tree grid. Type 1 toggles cell (y, x) between empty/tree; type 2 counts trees in an inclusive query rectangle. **In:** n q ; grid; then 1 y x or 2 y_1 x_1 y_2 x_2
Out: tree count for each type-2 query
Constraints: $n \leq 1000$, $q \leq 2 \cdot 10^5$.

1749 — List Removals

Start with list $x_1, \dots, x_n$. Repeatedly remove and print the element currently at position p_i ; positions are relative to the shrinking list. **In:** n ; values $x_1 \dots x_n$; removals $p_1 \dots p_n$
Out: removed values in order
Constraints: $n \leq 2 \cdot 10^3$; $x_i \leq 10^9$; $1 \leq p_i \leq n - i + 1$.

2166 — Prefix Sum Queries

Maintain an array under point assignments. For a query range $[a, b]$, report the maximum prefix sum $\max(0, x_a, x_a + x_{a+1}, \dots, \sum_{i=a}^b x_i)$. **In:** n q ; array; 1 k u (set) or 2 a b (query)
Out: maximum prefix sum for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^5$; $|x_i|, |u| \leq 10^9$. Use 64-bit.

2184 — Missing Coin Sum Queries

Coins have positive values x_i . For each index range $[a, b]$, using each coin there at most once, find the smallest positive sum that cannot be formed. **In:** n q ; coin values; then q lines a b
Out: smallest nonrepresentable positive sum for each query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^9$. Use 64-bit.

2206 — Pizzeria Queries

Building i sells pizza for p_i . Delivery from pizzeria a to customer at k costs $p_a + |a - k|$. Support point price changes and queries for the minimum delivered price at building k . **In:** n q ; prices; then 1 k x or 2 k
Out: $\min_a (p_a + |a - k|)$ for each type-2 query
Constraints: $n, q \leq 2 \cdot 10^5$; $p_i, x \leq 10^9$. Use 64-bit.

2416 — Increasing Array Queries

For each subarray $x_a, \dots, x_b$, you may only increment elements. Find the minimum total number of $+1$ operations needed to make it nondecreasing ($x_i \geq x_{i-1}$). **In:** n q ; array; then q lines a b
Out: minimum increments for each query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i \leq 10^9$. Use 64-bit.

3163 — Range Interval Queries

Given fixed $x_1, \dots, x_n$, each query (a, b, c, d) asks how many indices satisfy both $a \leq i \leq b$ and $c \leq x_i \leq d$. **In:** n q ; array; then a b c d
Out: matching index count for each query
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i, c, d \leq 10^9$.

3226 — Subarray Sum Queries II

Given a fixed integer array, for each range $[a, b]$ report the maximum subarray sum contained entirely inside that range. Empty subarrays with sum 0 are allowed. **In:** n q ; array; then q lines a b
Out: maximum subarray sum in each query range
Constraints: $n, q \leq 2 \cdot 10^5$; $|x_i| \leq 10^9$. Use 64-bit.

3304 — Visible Buildings Queries

For each query $[a, b]$, pretend only those buildings exist and view them from the left. Count buildings whose height is *strictly greater* than every earlier height in the range. **In:** n q ; heights; then q lines a b
Out: number of visible buildings per query
Constraints: $n \leq 10^5$, $q \leq 2 \cdot 10^5$; $h_i \leq 10^9$.

3356 — Distinct Values Queries II

Maintain an array under point assignments. For each range query $[a, b]$, determine whether every value in the range is pairwise distinct. **In:** n q ; array; then 1 k u or 2 a b
Out: YES if all values in $[a, b]$ are distinct, otherwise NO
Constraints: $n, q \leq 2 \cdot 10^5$; $x_i, u \leq 10^9$.